

The Grassroots-to-Mastery Roadmap

Phase 4: Special Phase Conversions & Asymmetrical Matrix Modeling

The Scott-T Connection, MMF Balance Equations, and Unbalanced Two-Phase Load Reflection

Module Objective:

To master the structural transformation from a balanced 3-phase infinite busbar to an isolated 2-phase distribution network. You will mathematically derive the critical 86.6% altitude tap, prove the "Balanced Load Theorem" demonstrating symmetrical grid extraction from 2-phase loads, and perform exhaustive complex phasor analyses to compute line currents under severely asymmetric power factors.

Pedagogical Sequence Framework

Phase 4 of 6

May 22, 2026

Contents

1	The Necessity of Phase Conversion	3
1.1	Introduction to the Scott-T Topology	3
2	Geometric Derivation of the Teaser Tap Ratio	3
2.1	The Equilateral Voltage Triangle	4
2.2	Deriving the Turn Ratio Asymmetry	5
3	MMF Balance and Primary Line Current Mapping	6
3.1	Current in Line A ($\vec{I}_A$)	6
3.2	Current in Lines B and C ($\vec{I}_B$ and $\vec{I}_C$)	6
4	The Balanced Load Theorem	7
4.1	Assuming a Balanced 2-Phase Condition	8
4.2	Evaluating Primary Line A	8
4.3	Evaluating Primary Line B	8
4.4	Evaluating Primary Line C	9
5	Unbalanced Load Asymmetry: Deep-Dive Case Study	9
5.1	Phase 1: Establishing the Coordinate Space and Secondary Phasors	9
5.2	Phase 2: Calculating Primary Line A	10
5.3	Phase 3: Calculating Reflected Main Current ($\vec{I}_M$)	10
5.4	Phase 4: Resolving Primary Line B	10
5.5	Phase 5: Resolving Primary Line C	10
6	Component Utilization & Rating Economics	11
6.1	The KVA Rating Penalty	11
7	Phase 4 Review and Self-Evaluation	12
7.1	Theoretical Competency Checklist	12

1 The Necessity of Phase Conversion

Cognitive Thought Process & Critical Insight

1. Why do we need to convert 3-phase power to 2-phase power? The national grid generates and transmits exclusively in 3-phase.
2. Certain legacy industrial loads and specific modern applications (like electric arc furnaces, robotic welding, and alternating-current railway traction networks) require 2-phase power (90° phase displacement).
3. If we simply tap two phases of a 3-phase grid to run a single-phase or 2-phase load, we severely unbalance the generator. We need a topology that draws balanced 3-phase power from the grid while delivering isolated 2-phase power. Enter the Scott Connection.

1.1 Introduction to the Scott-T Topology

Invented by Charles F. Scott in the late 19th century, the Scott connection utilizes two distinct, specialized single-phase transformers to mathematically map a 3-phase coordinate system into a 2-phase orthogonal coordinate system.

The two single-phase transformers are designated as:

- **The Main Transformer:** Connected directly across two lines of the 3-phase supply (e.g., Lines B and C). It must feature a center tap located at exactly 50% of its primary winding.
- **The Teaser Transformer:** Connected between the remaining 3-phase line (Line A) and the 50% center tap of the Main transformer.

2 Geometric Derivation of the Teaser Tap Ratio

MANDATORY MULTIMEDIA INTEGRATION: YOUTUBE LECTURE

Video Topic: Scott Connection Phasor Geometry and Tap Derivation

Search Query / Link: https://www.youtube.com/results?search_query=scott+connection+transformer+phasor+diagram

Required Focus: Focus on how the altitude of the equilateral voltage triangle dictates the required number of turns on the Teaser primary winding.

Cognitive Thought Process & Critical Insight

1. A transformer's voltage ratio is strictly proportional to its turns ratio ($V \propto N$).
2. The Main transformer sits across the full line-to-line voltage (V_L). Let it have N_1 primary turns.
3. The Teaser transformer sits between a line terminal and the center of the opposite phase. We must use trigonometry to find the exact voltage impressed across the Teaser primary.
4. We will prove that the voltage across the Teaser is exactly $\frac{\sqrt{3}}{2}V_L$, requiring the primary winding to be tapped at 86.6%.

2.1 The Equilateral Voltage Triangle

Assume a balanced 3-phase supply with a phase sequence A-B-C and a line-to-line voltage magnitude of V_L . The line voltage phasors form a perfect equilateral triangle.

Let us assign the voltage across the Main transformer (Lines B and C) as our reference phasor along the horizontal axis:

$$\vec{V}_{BC} = V_L \angle 0^\circ \quad (1)$$

The potentials of terminals B and C with respect to the neutral point can be derived from the line voltage. However, the Main transformer has a center tap, let's call it N_{tap} . Because N_{tap} is exactly at the midpoint of winding B-C, its potential lies precisely halfway between B and C.

If we place the origin of our coordinate system at the center tap N_{tap} :

$$\vec{V}_{B,N_{\text{tap}}} = -\frac{V_L}{2}, \quad \vec{V}_{C,N_{\text{tap}}} = +\frac{V_L}{2} \quad (2)$$

The Teaser transformer is connected between line terminal A and N_{tap} . In an equilateral triangle of side length V_L , the line connecting an apex (A) to the midpoint of the opposite base (B-C) is the **altitude (median)** of the triangle.

Using basic trigonometry or the Pythagorean theorem on the right-angled triangle formed by A, B, and N_{tap} :

$$(V_{AB})^2 = (V_{A,N_{\text{tap}}})^2 + (V_{B,N_{\text{tap}}})^2 \quad (3)$$

$$(V_L)^2 = (V_{A,N_{\text{tap}}})^2 + \left(\frac{V_L}{2}\right)^2 \quad (4)$$

$$(V_{A,N_{\text{tap}}})^2 = V_L^2 - \frac{V_L^2}{4} = \frac{3}{4}V_L^2 \quad (5)$$

$$V_{A,N_{\text{tap}}} = \frac{\sqrt{3}}{2}V_L \approx 0.866V_L \quad (6)$$

Because the altitude is perpendicular to the base, the phasor $\vec{V}_{A,N_{\text{tap}}}$ is shifted exactly 90° relative to $\vec{V}_{BC}$:

$$\vec{V}_{\text{Teaser Primary}} = \frac{\sqrt{3}}{2}V_L \angle 90^\circ \quad (7)$$

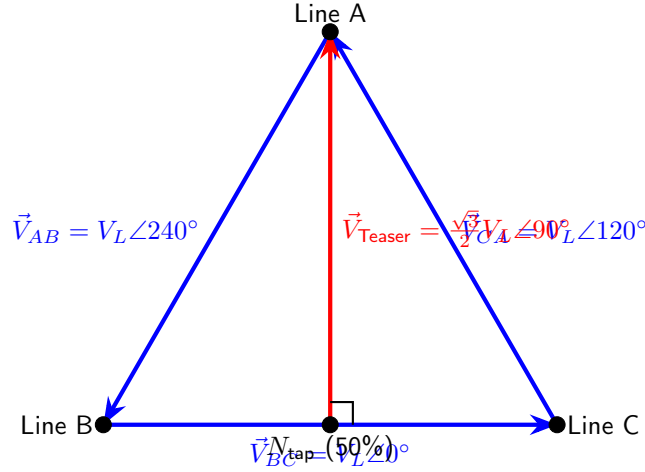


Figure 1: The Primary Voltage Phasor Triangle. The Teaser primary voltage is the altitude of the 3-phase equilateral triangle.

2.2 Deriving the Turn Ratio Asymmetry

To supply a balanced 2-phase secondary system, the magnitudes of the induced voltages on the secondary windings of both the Main and Teaser transformers must be strictly equal:

$$|V_{2,\text{Main}}| = |V_{2,\text{Teaser}}| = V_2 \quad (8)$$

The Main transformer has the full line voltage V_L applied across N_1 primary turns. Assuming N_2 secondary turns:

$$V_{2,\text{Main}} = \frac{N_2}{N_1} V_L \quad (9)$$

The Teaser transformer has only $\frac{\sqrt{3}}{2} V_L$ applied across its primary. To achieve the exact same secondary voltage V_2 using the identical secondary coil of N_2 turns, we must adjust its primary turns (N_T):

$$V_{2,\text{Teaser}} = \frac{N_2}{N_T} \left(\frac{\sqrt{3}}{2} V_L \right) \quad (10)$$

Setting $V_{2,\text{Main}} = V_{2,\text{Teaser}}$:

$$\frac{N_2}{N_1} V_L = \frac{N_2}{N_T} \left(\frac{\sqrt{3}}{2} V_L \right) \quad (11)$$

$$\frac{1}{N_1} = \frac{\sqrt{3}}{2N_T} \implies N_T = \frac{\sqrt{3}}{2} N_1 \approx 0.866 N_1 \quad (12)$$

Engineering Conclusion: The Teaser primary winding must be designed with (or tapped at) exactly 86.6% of the total turns of the Main primary winding to output equal secondary voltages.

3 MMF Balance and Primary Line Current Mapping

Cognitive Thought Process & Critical Insight

1. Now that we have established the voltage vectors, we must analyze the currents.
2. Power flows from the 2-phase secondary back to the 3-phase primary. We use the law of Conservation of Magnetomotive Force (Ampere-turns balance) for each core independently.
3. The Teaser core dictates the current entering Line A.
4. The Main core dictates the currents entering Lines B and C, but remember: the Teaser return current splits through the center tap and flows through the Main primary winding!

Let the two independent secondary loads draw currents $\vec{I}_{2M}$ (from the Main secondary) and $\vec{I}_{2T}$ (from the Teaser secondary).

3.1 Current in Line A ($\vec{I}_A$)

Line A connects directly to the top of the Teaser primary winding. By balancing the MMF across the Teaser core (ignoring the small no-load magnetizing current):

$$\vec{I}_A N_T = \vec{I}_{2T} N_2 \quad (13)$$

Substitute $N_T = \frac{\sqrt{3}}{2} N_1$:

$$\vec{I}_A \left(\frac{\sqrt{3}}{2} N_1 \right) = \vec{I}_{2T} N_2 \quad (14)$$

Let $n = N_1/N_2$ be the base transformation ratio of the main core.

$$\vec{I}_A = \frac{2}{\sqrt{3}} \frac{N_2}{N_1} \vec{I}_{2T} = \frac{2}{\sqrt{3}n} \vec{I}_{2T} \approx 1.15 \frac{\vec{I}_{2T}}{n} \quad (15)$$

3.2 Current in Lines B and C ($\vec{I}_B$ and $\vec{I}_C$)

The analysis for the Main transformer is more complex because its primary winding handles two distinct MMF sources:

1. The current reflected strictly by its own secondary load ($\vec{I}_{2M}$). We will call this pure reflected component $\vec{I}_M$.

$$\vec{I}_M = \frac{N_2}{N_1} \vec{I}_{2M} = \frac{\vec{I}_{2M}}{n} \quad (16)$$

2. The returning current from the Teaser primary. Current $\vec{I}_A$ enters the Teaser, flows to the center tap N_{tap} on the Main winding, and must split into two equal halves to return to the grid via Lines B and C. By KCL at the center tap, half of $\vec{I}_A$ flows towards B, and half flows towards C. Because they flow from the center outward, they flow in opposite directions relative to the main loop path.

Superimposing these two components, we write the rigorous node equations for the line currents at terminals B and C:

$$\vec{I}_B = \vec{I}_M - \frac{\vec{I}_A}{2} = \frac{\vec{I}_{2M}}{n} - 0.5\vec{I}_A \quad (17)$$

$$\vec{I}_C = -\vec{I}_M - \frac{\vec{I}_A}{2} = -\frac{\vec{I}_{2M}}{n} - 0.5\vec{I}_A \quad (18)$$

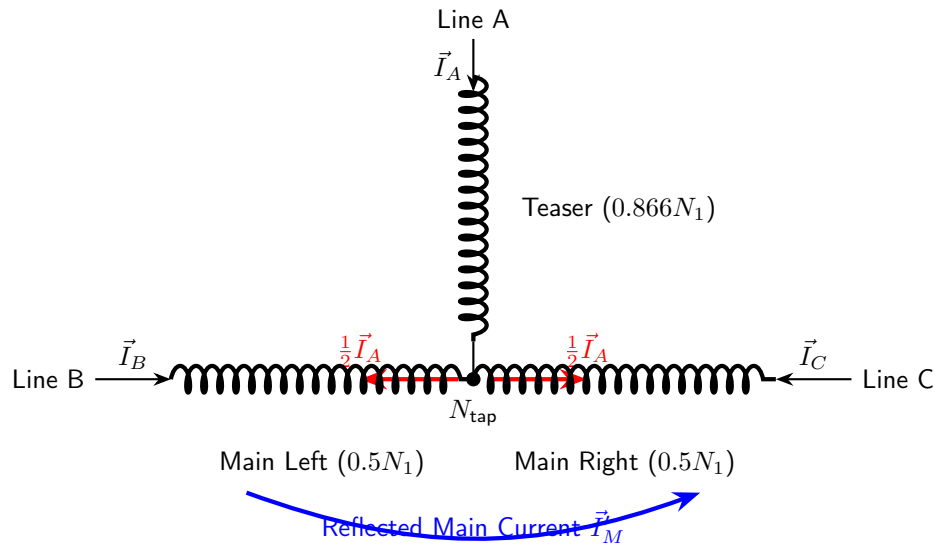


Figure 2: MMF Schematic. Shows how the Teaser current splits at the center tap, superimposing on the internally circulating reflected Main current.

4 The Balanced Load Theorem

Cognitive Thought Process & Critical Insight

1. A crucial claim of the Scott connection is that if you place a perfectly balanced 2-phase load on the secondary, it will draw a perfectly balanced 3-phase load from the primary grid.
2. We must prove this mathematically. A balanced 3-phase line means $|\vec{I}_A| = |\vec{I}_B| = |\vec{I}_C|$ and they are exactly 120° apart.
3. Let's substitute balanced 2-phase constraints ($I_{2M} = I_{2T}$, same power factor) into our MMF equations and evaluate the vector results.

4.1 Assuming a Balanced 2-Phase Condition

Let the secondary loads be equal in magnitude and power factor. Let I_2 be the secondary current magnitude and ϕ be the lagging phase angle for both units.

Taking the Main secondary voltage $\vec{V}_{2M}$ as the reference horizontal phasor (0°):

$$\vec{V}_{2M} = V_2 \angle 0^\circ \implies \vec{I}_{2M} = I_2 \angle -\phi = I_2(\cos \phi - j \sin \phi) \quad (19)$$

The Teaser secondary voltage is orthogonal (90°):

$$\vec{V}_{2T} = V_2 \angle 90^\circ \implies \vec{I}_{2T} = I_2 \angle (90^\circ - \phi) = I_2(\sin \phi + j \cos \phi) \quad (20)$$

4.2 Evaluating Primary Line A

$$\vec{I}_A = \frac{2}{\sqrt{3}n} \vec{I}_{2T} = \frac{2I_2}{\sqrt{3}n} (\sin \phi + j \cos \phi) \quad (21)$$

For simplicity in comparison, let $\frac{2I_2}{\sqrt{3}n} = K$. Then:

$$\vec{I}_A = K \sin \phi + jK \cos \phi \quad (22)$$

The magnitude is $|\vec{I}_A| = \sqrt{(K \sin \phi)^2 + (K \cos \phi)^2} = K$.

4.3 Evaluating Primary Line B

First, find $\vec{I}_M$:

$$\vec{I}_M = \frac{\vec{I}_{2M}}{n} = \frac{I_2}{n} (\cos \phi - j \sin \phi) \quad (23)$$

Notice that $\frac{I_2}{n}$ can be expressed in terms of K (since $K = \frac{2I_2}{\sqrt{3}n} \implies \frac{I_2}{n} = \frac{\sqrt{3}}{2}K$).

$$\vec{I}_M = K \frac{\sqrt{3}}{2} \cos \phi - jK \frac{\sqrt{3}}{2} \sin \phi \quad (24)$$

Now apply the node equation for $\vec{I}_B = \vec{I}_M - 0.5\vec{I}_A$:

$$\vec{I}_B = \left(K \frac{\sqrt{3}}{2} \cos \phi - jK \frac{\sqrt{3}}{2} \sin \phi \right) - 0.5 (K \sin \phi + jK \cos \phi) \quad (25)$$

$$= K \left[\left(\frac{\sqrt{3}}{2} \cos \phi - \frac{1}{2} \sin \phi \right) + j \left(-\frac{\sqrt{3}}{2} \sin \phi - \frac{1}{2} \cos \phi \right) \right] \quad (26)$$

Using angle subtraction identities ($\cos(\phi + 30^\circ) = \cos \phi \cos 30^\circ - \sin \phi \sin 30^\circ$):

$$\vec{I}_B = K [\cos(\phi + 30^\circ) - j \sin(\phi + 30^\circ)] = K \angle -(\phi + 30^\circ) \quad (27)$$

The magnitude is $|\vec{I}_B| = K$.

4.4 Evaluating Primary Line C

Apply the node equation $\vec{I}_C = -\vec{I}_M - 0.5\vec{I}_A$:

$$\vec{I}_C = \left(-K \frac{\sqrt{3}}{2} \cos \phi + jK \frac{\sqrt{3}}{2} \sin \phi \right) - 0.5 (K \sin \phi + jK \cos \phi) \quad (28)$$

$$= K \left[\left(-\frac{\sqrt{3}}{2} \cos \phi - \frac{1}{2} \sin \phi \right) + j \left(\frac{\sqrt{3}}{2} \sin \phi - \frac{1}{2} \cos \phi \right) \right] \quad (29)$$

Using angle addition identities, this reduces to:

$$\vec{I}_C = K \angle (-\phi + 210^\circ) \equiv K \angle -(\phi - 210^\circ) \equiv K \angle -(\phi + 150^\circ) \quad (30)$$

The magnitude is $|\vec{I}_C| = K$.

Q.E.D. (Quod Erat Demonstrandum): The magnitudes are identical ($|\vec{I}_A| = |\vec{I}_B| = |\vec{I}_C| = K$). The phase angles are displaced by exactly 120° and 240° . A balanced 2-phase load rigorously reflects as a perfectly balanced 3-phase load on the primary grid.

5 Unbalanced Load Asymmetry: Deep-Dive Case Study

Cognitive Thought Process & Critical Insight

1. In university exams, you will almost never be given a balanced load. You will be given disjointed power factors (one leading, one lagging).
2. We must apply the MMF equations strictly using complex rectangular algebra. Polar math often fails here because addition/subtraction in polar coordinates is error-prone.
3. We will solve an advanced constraint: 25 kW leading at 0.75 on the Teaser, and 45 kW lagging at 0.75 on the Main, from a 6600V primary to a 220V secondary.

5.1 Phase 1: Establishing the Coordinate Space and Secondary Phasors

Given Parameters:

- $V_1 = 6600\text{V}, V_2 = 220\text{V} \implies n = \frac{6600}{220} = 30$
- Power Factor $\cos \phi = 0.75 \implies \phi = 41.4096^\circ$.
- Leading means we ADD the angle to the voltage phase. Lagging means we SUBTRACT.

Main Secondary Current (I_{2M}):

$$P_M = 45000\text{W} \implies I_{2M} = \frac{45000}{220 \times 0.75} = 272.727 \text{ A}$$

Base angle is 0° . Current is lagging.

$$\vec{I}_{2M} = 272.727 \angle -41.41^\circ \text{ A} = 204.54 - j180.36 \text{ A}$$

Teaser Secondary Current (I_{2T}):

$$P_T = 25000 \text{ W} \implies I_{2T} = \frac{25000}{220 \times 0.75} = 151.515 \text{ A}$$

Base angle is 90° . Current is leading.

$$\vec{I}_{2T} = 151.515 \angle (90^\circ + 41.41^\circ) = 151.515 \angle 131.41^\circ \text{ A} = -100.22 + j113.63 \text{ A}$$

5.2 Phase 2: Calculating Primary Line A

$$\vec{I}_A = \frac{2}{\sqrt{3}n} \vec{I}_{2T} = \frac{2}{\sqrt{3} \times 30} (-100.22 + j113.63)$$

$$\frac{2}{51.96} = 0.03849$$

$$\vec{I}_A = 0.03849 \times (-100.22 + j113.63) = -\mathbf{3.857} + \mathbf{j4.374} \text{ A}$$

Magnitude: $|\vec{I}_A| = 5.832 \text{ A}$.

5.3 Phase 3: Calculating Reflected Main Current ($\vec{I}'_M$)

$$\vec{I}'_M = \frac{\vec{I}_{2M}}{n} = \frac{204.54 - j180.36}{30} = \mathbf{6.818} - \mathbf{j6.012} \text{ A}$$

5.4 Phase 4: Resolving Primary Line B

$$\vec{I}_B = \vec{I}'_M - 0.5\vec{I}_A$$

$$0.5\vec{I}_A = -1.9285 + j2.187 \text{ A}$$

$$\vec{I}_B = (6.818 - j6.012) - (-1.9285 + j2.187)$$

$$\vec{I}_B = (6.818 + 1.9285) + j(-6.012 - 2.187) = \mathbf{8.7465} - \mathbf{j8.199} \text{ A}$$

Magnitude: $|\vec{I}_B| = \sqrt{8.7465^2 + (-8.199)^2} = \mathbf{11.989} \text{ A}$. Angle: $-\mathbf{43.14}^\circ$.

5.5 Phase 5: Resolving Primary Line C

$$\vec{I}_C = -\vec{I}'_M - 0.5\vec{I}_A$$

$$\vec{I}_C = (-6.818 + j6.012) - (-1.9285 + j2.187)$$

$$\vec{I}_C = (-6.818 + 1.9285) + j(6.012 - 2.187) = -\mathbf{4.8895} + \mathbf{j3.825} \text{ A}$$

Magnitude: $|\vec{I}_C| = \sqrt{(-4.8895)^2 + 3.825^2} = \mathbf{6.208} \text{ A}$. Angle: $\mathbf{141.9}^\circ$.

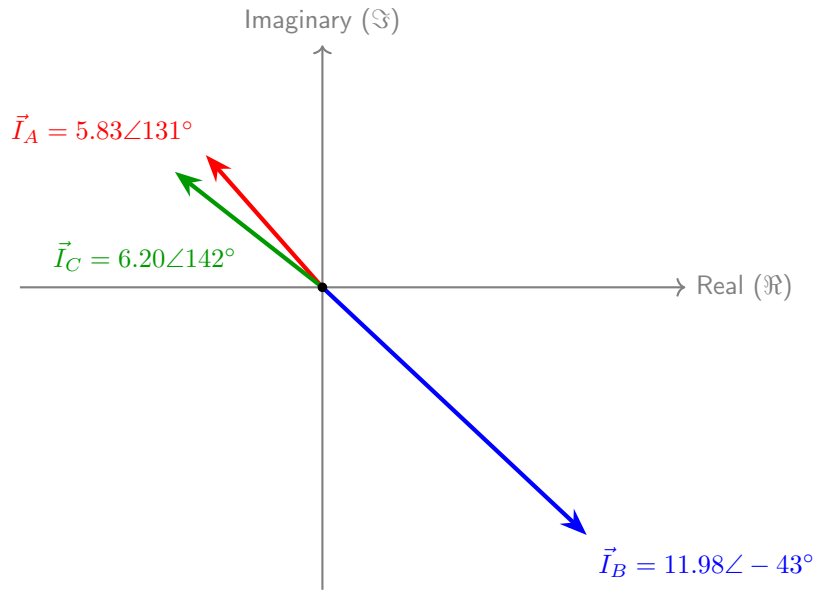


Figure 3: Complex Plane Plot of Asymmetrical Primary Line Currents. Notice the severe imbalance caused by differing power factors.

6 Component Utilization & Rating Economics

Cognitive Thought Process & Critical Insight

1. Does the Scott connection utilize 100% of its copper? No.
2. The Main transformer has a center tap where currents flow in opposite directions, and the Teaser handles $\sqrt{3}/2$ voltage but full line current.
3. We must calculate the Utilization Factor to understand the economic penalty of using a Scott connection versus standard 3-phase banks.

6.1 The KVA Rating Penalty

Assume the network is delivering a perfectly balanced 2-phase load. Let the total active power delivered to the secondary be $P = 2V_2I_2$ at unity power factor.

Teaser Transformer Rating: The Teaser primary winding has $\frac{\sqrt{3}}{2}V_L$ impressed across it, and it carries the primary line current I_L .

$$S_{\text{Teaser}} = \left(\frac{\sqrt{3}}{2}V_L \right) I_L = 0.866V_L I_L \quad (31)$$

Main Transformer Rating: The Main primary winding has the full V_L impressed across it, and it carries the line current I_L . Even though the current splits at the center tap, the copper

cross-section must be rated for the full line current continuously.

$$S_{\text{Main}} = V_L I_L \quad (32)$$

Total Transformer Bank KVA:

$$S_{\text{Total}} = S_{\text{Teaser}} + S_{\text{Main}} = 0.866V_L I_L + V_L I_L = 1.866V_L I_L \quad (33)$$

Active Power Output vs Equivalent 3-Phase Input: The total power output drawn from the 3-phase grid is $P_{\text{grid}} = \sqrt{3}V_L I_L = 1.732V_L I_L$.

Utilization Factor:

$$\text{Utilization Factor} = \frac{\text{Total Active Power}}{\text{Total Installed KVA Rating}} = \frac{1.732V_L I_L}{1.866V_L I_L} = \mathbf{0.928} \text{ (or 92.8\%)} \quad (34)$$

This means that to supply 100 kVA of 2-phase load, you must purchase and install ****107.7 kVA**** worth of physical transformer copper and iron. There is a 7.7% economic penalty built into the Scott connection due to asymmetrical current routing and inactive copper segments.

7 Phase 4 Review and Self-Evaluation

Cognitive Thought Process & Critical Insight

Verify your engineering competency profile against the technical checklist below before advancing to Phase 5 (Dynamic Regulation Systems).

7.1 Theoretical Competency Checklist

1. **Altitude Derivation:** Can you use Pythagorean geometry to prove why the Teaser transformer must be tapped at exactly 86.6% of the Main transformer's winding?
2. **MMF Equations:** Write down the three nodal MMF equations for the primary line currents ($\vec{I}_A$, $\vec{I}_B$, $\vec{I}_C$) based on secondary loads $\vec{I}_{2T}$ and $\vec{I}_{2M}$.
3. **Symmetry Theorem:** If asked to prove that a balanced 2-phase load yields a balanced 3-phase input, can you perform the trigonometric reduction to show $|\vec{I}_A| = |\vec{I}_B| = |\vec{I}_C|$?
4. **Asymmetric Analysis:** In a complex problem with leading and lagging power factors, why must you convert all polar phasors to rectangular forms before applying the node equations?
5. **Utilization Economics:** What is the utilization factor of a Scott connection, and why does the Main transformer require more installed KVA than the power it actively transfers?

You have successfully completed Phase 4. You have mastered Scott connection topology, MMF balancing, and advanced asymmetric load reflections. You are now fully prepared to advance to Phase 5: Dynamic Regulation Systems and On-Load Tap Changers (OLTC).